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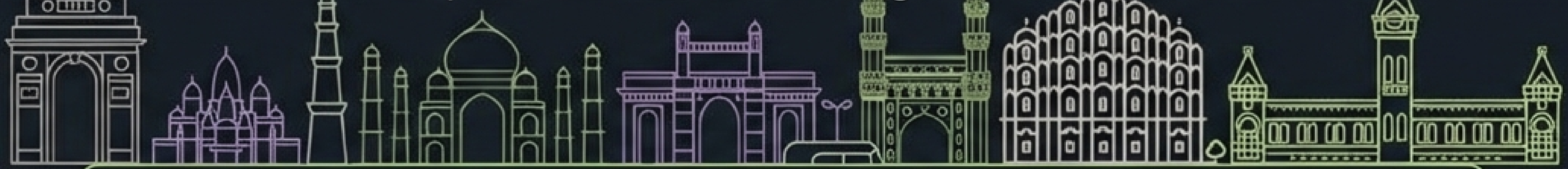
AI for Bharat Hackathon

Green Sentinel

Developers: Shraddha Shravagi & Sudhindra Kini



Protecting **₹18 lakh crore** in annual Indian crop value by mitigating **₹90,000 crore** in preventable losses caused by late-detected diseases, fires, and wildlife intrusions.

The Digital Immune System for Indian Agriculture

Green Sentinel is a production-deployed, cloud-native agricultural intelligence platform built for the realities of Bharat: Mobile-first (375px viewport), Offline-first (PWA), and WhatsApp-first.

Satellite Analytics

Weekly crop health scores without field visits using free Sentinel-2 NDVI data.

Edge AI Vision

Autonomous threat surveillance turning cheap CCTV into AI guards.

AI Disease Forecasting

Predicting and diagnosing crop diseases in under 8 seconds.

World-Class Agricultural AI, Engineered for Bharat's Constraints

Why AI?	**It replaces the expensive agronomist.** Claude 3.5 Sonnet analyzes threats (fire, human intruders, nilgai) from cheap CCTV in seconds, bypassing the need for constant manual monitoring.
How AWS?	**100% Serverless architecture.** Powered by Lambda, DynamoDB, Amazon Bedrock, and API Gateway. We utilize a monolithic inline Lambda router with O(1) path-based dispatch to minimize cold starts and keep infrastructure pay-per-request.
The Value Added	**Immediate action despite language barriers.** Translates complex threat intelligence into regional languages (Hindi, Marathi, Tamil, etc.) via the Bhashini API, delivering instant WhatsApp audio and text alerts to farmers' phones.

Comprehensive Intelligence Pipeline

Satellite Health Tracking

NDVI, NDWI, and Moisture Index calculations via AWS Open Data Registry.

Weather Intelligence

Hyper-local, hourly conditions and 14-day forecasting via Open-Meteo.

Disease & Pest Forecasting

AI rule-based prediction alerting on Late Blight, Fall Armyworm, and Rust based on localized weather and crop stages.

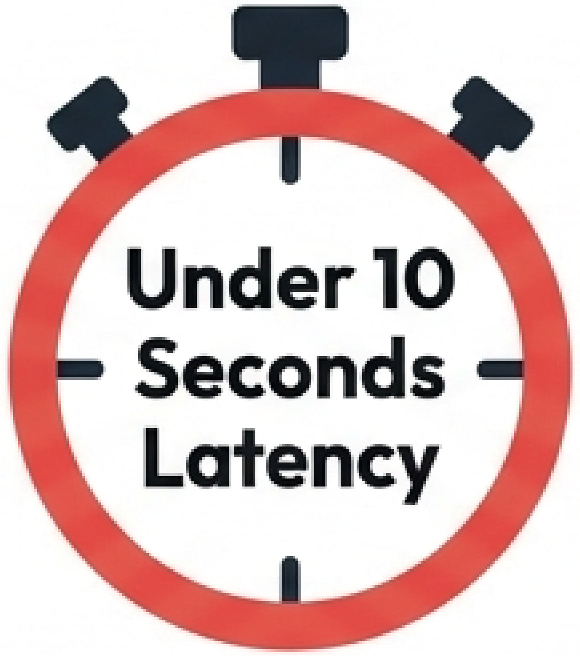
Smart Irrigation Modeling

Zone-based scheduling derived from Evapotranspiration (ET_0) and soil moisture estimates.

Yield Prediction

Machine learning models factoring historical data and current vegetation indices.

The <10 Second End-to-End Alert Pipeline



1. Capture
Edge Agent captures an RTSP camera frame every 5-10 seconds.

2. Analyze
Frame routed to Amazon Bedrock (Claude 3.5 Sonnet) for threat assessment (<3 second processing).

3. Process
AWS API Gateway & Lambda log the event in DynamoDB single-table schema.

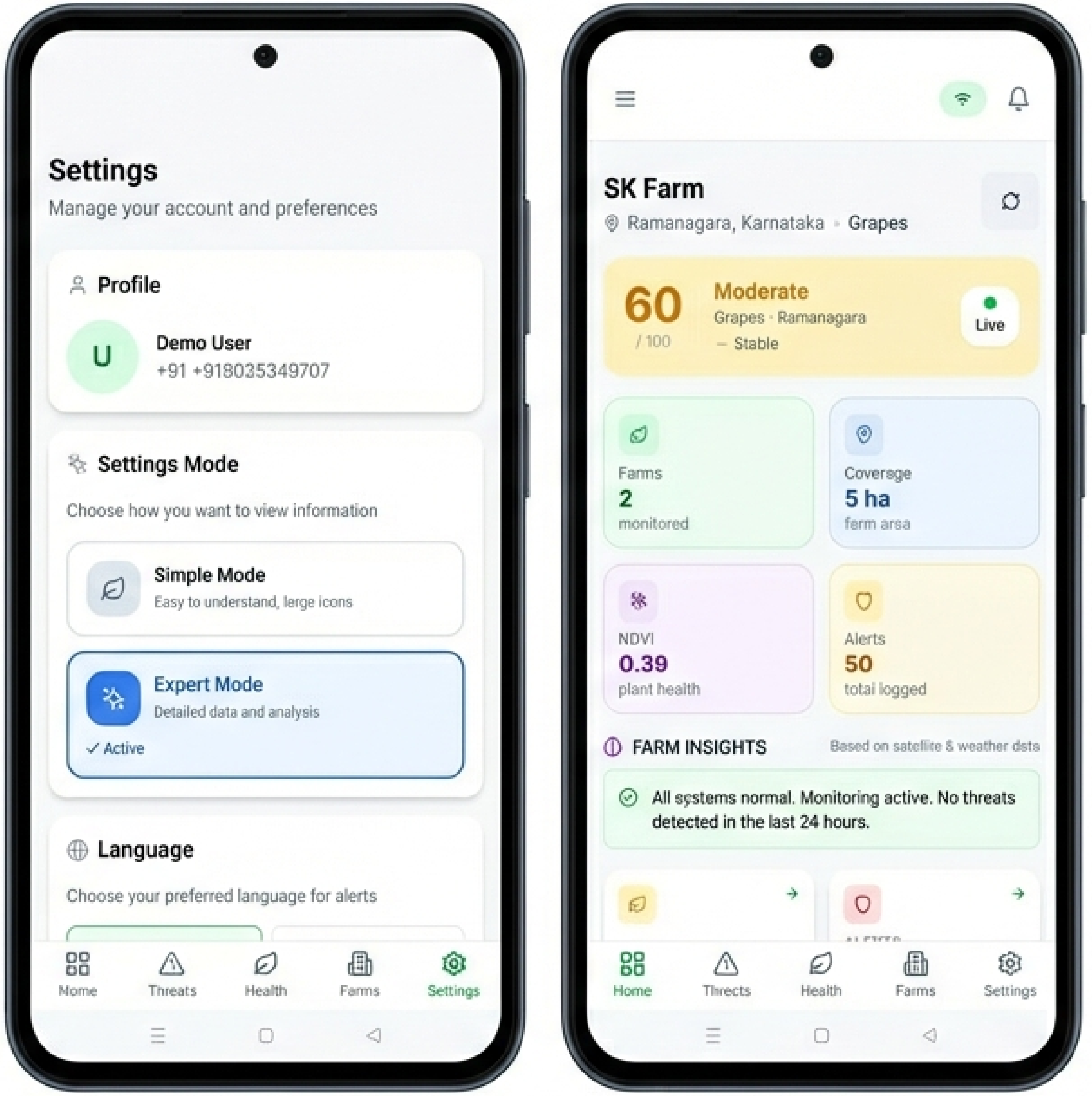
4. Translate
Bhashini API converts the alert into the farmer's native language.

5. Deliver
Twilio API pushes a high-priority WhatsApp message with photographic evidence.

UX Philosophy: Built for the 375px Viewport

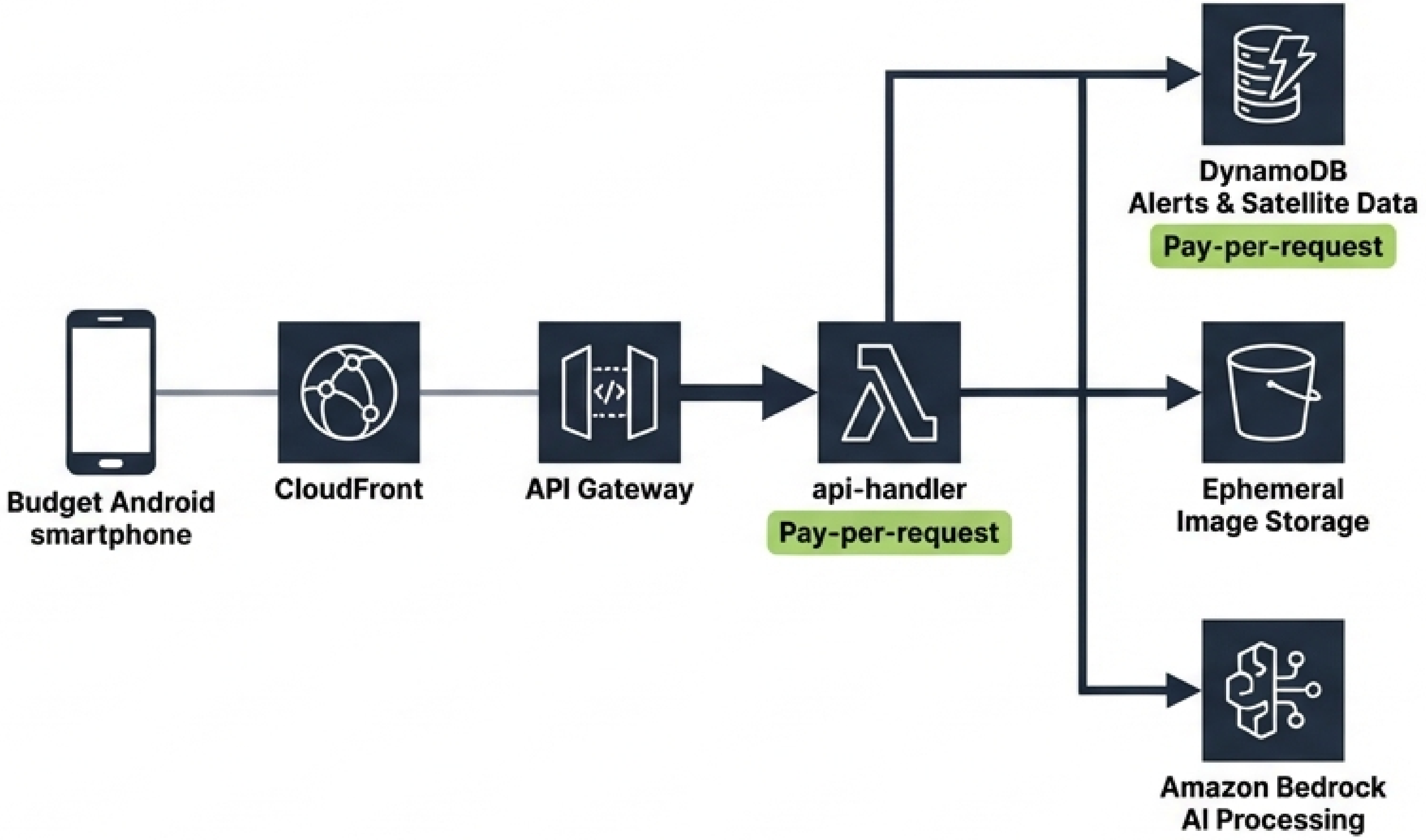
Our interface is a zero-install Progressive Web App (PWA) built specifically for budget Androids on fluctuating 2G/3G networks.

- **Offline-First Architecture:** Zustand stores persist to localStorage; Service Workers cache the UI and 7 days of satellite tiles.
- **Adaptive Complexity:** A toggle between Simple Mode (large icons, basic metrics) and Expert Mode (detailed charts, NDWI data) accommodates different literacy levels.



Cloud-Native Serverless Topology

- **Frontend**: CloudFront CDN hosting the React 18 / Vite PWA.
- **Compute**: Monolithic api-handler Lambda in Node.js 20.x. Deliberate choice: one cold start pool, 8.7 KB inline code, zero zip dependencies.
- **Storage & State**: DynamoDB using PAY_PER_REQUEST single-table design for alerts and satellite data. S3 for ephemeral image storage (30-day lifecycle).
- **Orchestration**: EventBridge triggering sequential satellite-processor pipelines.



The Technology Stack

Frontend Layer	React 18, Vite, TypeScript 5.3, Tailwind CSS, Zustand (persisted state), Workbox PWA.
Backend & AWS Layer	Node.js 20.x, AWS SDK v3, AWS CDK (Infrastructure as Code), AWS Cognito (Phone OTP Auth).
AI & Intelligence	Amazon Bedrock (Claude 3.5 Sonnet Vision), Bhashini (Translation & Voice Synthesis).
External Data APIs	**Element84** Earth Search (Free Sentinel-2 STAC API), Open-Meteo (Free weather, 10K requests/day), **Twilio** (WhatsApp Sandbox).

Radical Affordability: ~₹730/Month for a 3-Farm Operation

Every infrastructure rupee is justified through strict AWS Free Tier compliance and AI budget guards.

Compute: Lambda & Cognito (1M+ requests)	Free Tier
Database: DynamoDB (~50K ops)	~₹20
Storage & CDN: S3 & CloudFront (10GB egress)	~₹110
AI Vision & Alerts: Bedrock Threat Detection (6,000 Haiku + 100 Sonnet calls)	~₹230 ($1.80)

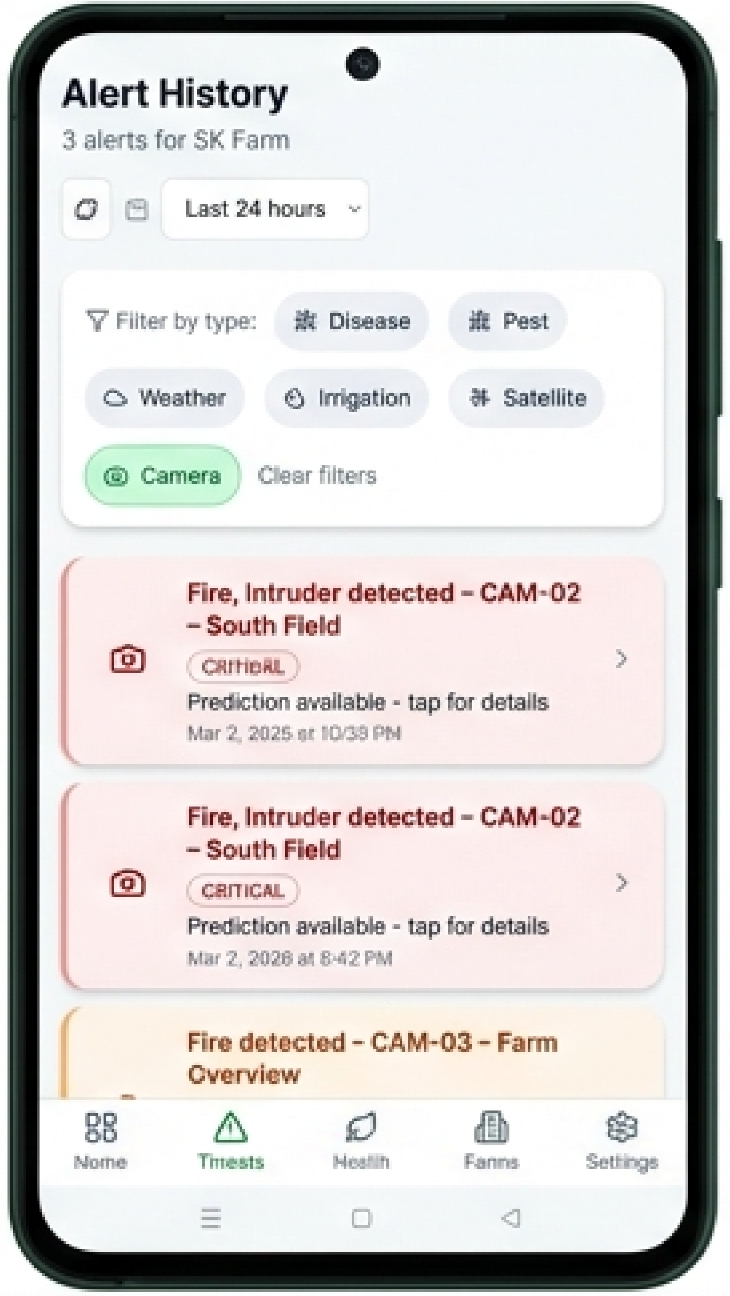
₹730/month

Strict Cost Guards: Server-side DynamoDB counters cut off API access after 200 daily AI calls to prevent runway costs.

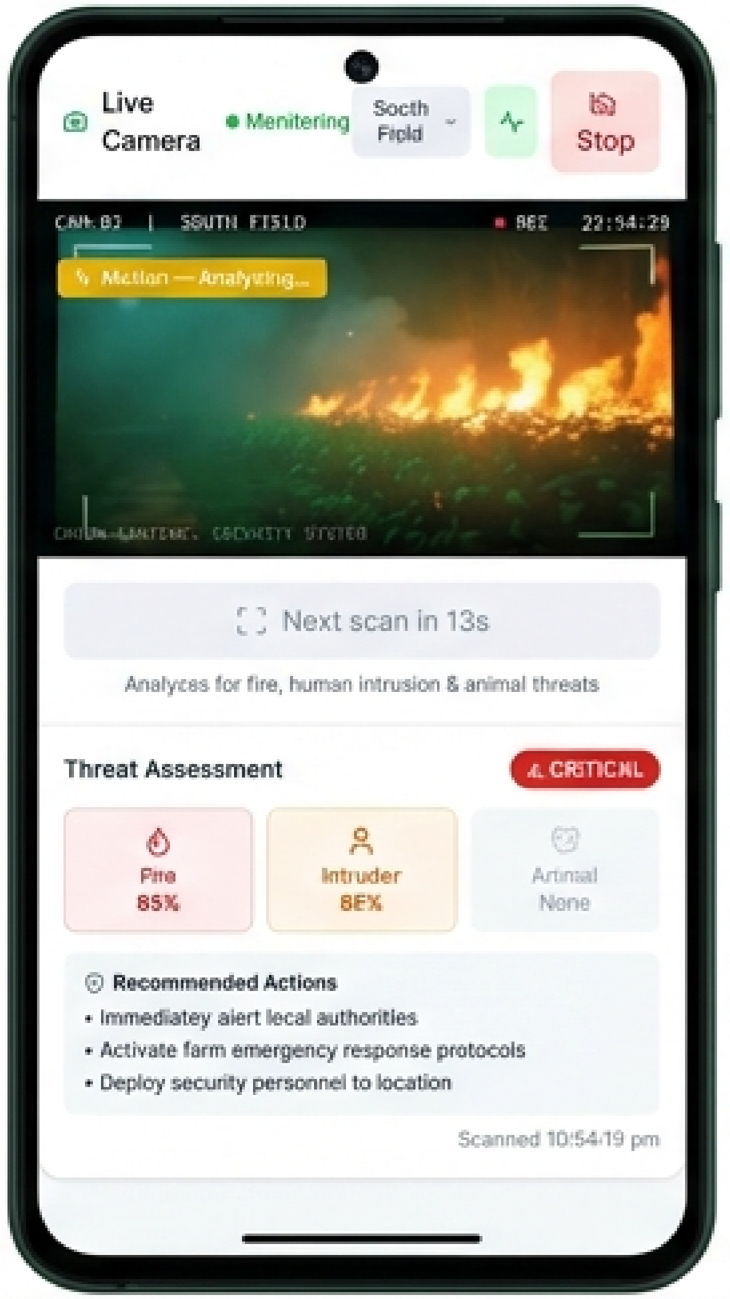
Prototype in Action: Threat Detection & Crop Health

Real-time surveillance detects threats autonomously, while satellite data provides continuous health benchmarking.

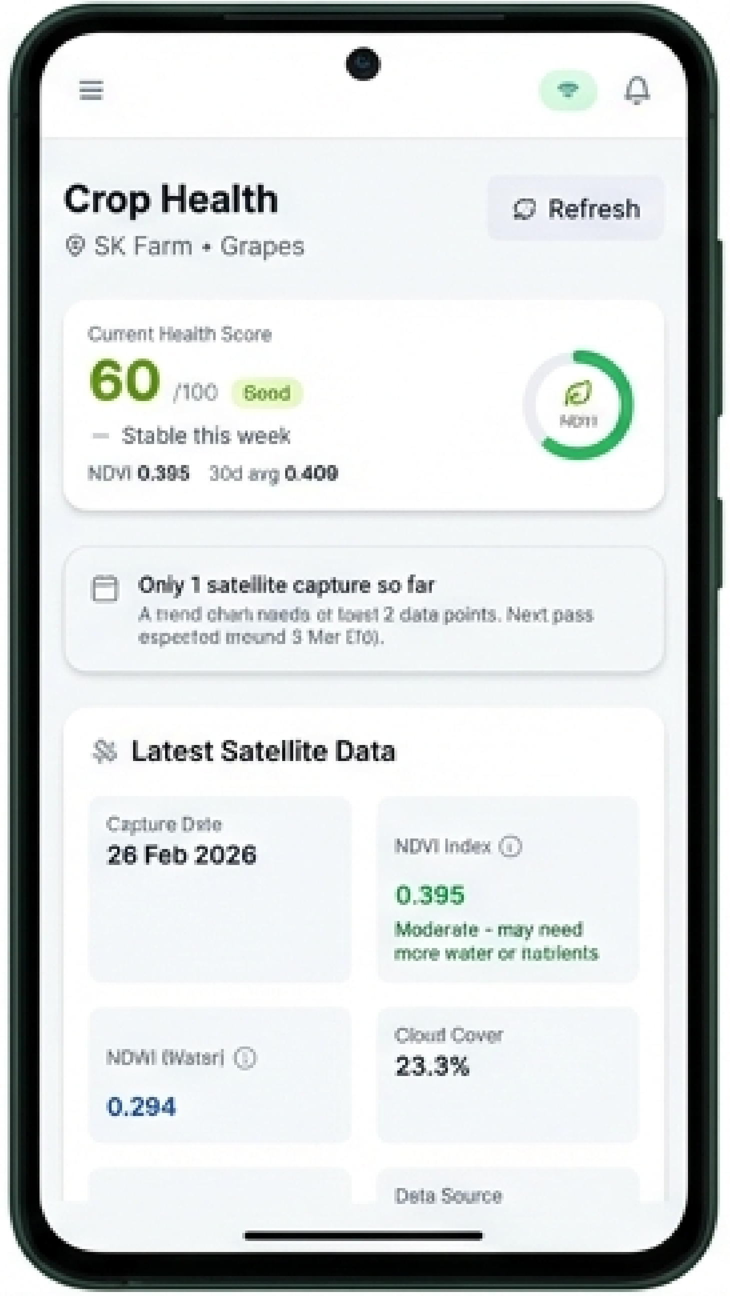
Critical Alerting: Identifying fires and human intruders with exact timestamps.



Live AI Camera: Autonomous threat detection powered by Claude Vision.



Satellite Crop Health: Sentinel-2 data distilled into an easy-to-read NDVI health score.



Performance & Benchmarking

Engineered for the reality of rural network latency and critical threat timelines.

<10s
End-to-End Latency
From frame capture on the farm to WhatsApp message delivery on the farmer's phone.

<3s
Frame Analysis
Amazon Bedrock Claude 3.5 Sonnet processing speed per image.

8s
PWA Load Time
Cold start initialization on standard 2G/3G networks.

99.5%
Target Uptime
Supported by exponential backoff retries on Lambda functions and S3 temporary caching.

The 3 Moats & Future Scale

Our Competitive Moats:

1. The Data Flywheel: Every scan and alert improves our localized Indian disease models (e.g., Late Blight on Nashik grapes).

2. Benign Hardware Lock-in: Once the Edge Agent runs on a farmer's existing CCTV, switching costs are physical, not just digital.

3. Deep Localization: Bypasses generic English AI chatbots by automatically 'seeing' the crop and acting in regional dialects.

Amazon SageMaker integration for advanced Yield Prediction modeling.

Multi-region expansion to Southeast Asia (Singapore) and Latin America via 1-click AWS CDK deploy.

Prototype Assets

GitHub Public Repository

Access our complete AWS CDK infrastructure, React frontend, and Edge Agent source code.

Demo Video

A 3-minute comprehensive walkthrough of the Green Sentinel Digital Immune System in action.

Green Sentinel

Empowering Bharat's farmers with the cloud.